Day 14 — Genome Mining

BIOL 4810

2026-03-08

Day 14 — Genome Mining (all tools required)

Goal: Use specialized web tools to find antibiotic resistance, virulence signals, predicted pathogenic potential, secondary metabolite gene clusters, prophages, genomic islands, and CRISPR-Cas systems in your genome assembly.
Input file: Your filtered **.fasta** from Galaxy.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample ID (use the same ID you use in file naming / Git):

Unk1

0B. Assembly file name (what you downloaded from Galaxy):

Unk1_SPAdes_filtered_contigs.fasta

0C. Suspected identity so far (from 16S/Prokka/TYGS, if you have one):

Salmonella enterica subsp. enterica serovar Typhimurium

Required websites (use these)

Open each tool in a new tab (links below open in a new tab automatically). We will demo how to submit FASTA/contigs to each tool.

- **CARD / RGI** (AMR): https://card.mcmaster.ca/analyze/rgi (https://card.mcmaster.ca/analyze/rgi)
- **ResFinder** (AMR genes, CGE): https://cge.food.dtu.dk/services/ResFinder/ (https://cge.food.dtu.dk/services/ResFinder/)
- **PathogenFinder 2** (pathogenic potential, CGE): https://cge.food.dtu.dk/services/PathogenFinder/ (https://cge.food.dtu.dk/services/PathogenFinder/)
- **VirulenceFinder** (virulence genes, CGE): https://cge.food.dtu.dk/services/VirulenceFinder/ (https://cge.food.dtu.dk/services/VirulenceFinder/)
- **antiSMASH** (secondary metabolite gene clusters): https://antismash.secondarymetabolites.org/ (https://antismash.secondarymetabolites.org/)
- **PHASTER** (prophages): https://phaster.ca/ (https://phaster.ca/)
- **IslandViewer** (genomic islands): https://www.pathogenomics.sfu.ca/islandviewer/ (https://www.pathogenomics.sfu.ca/islandviewer/)
- **CRISPRCasFinder** (CRISPR arrays + Cas genes): https://crisprcas.i2bc.paris-saclay.fr/ (https://crisprcas.i2bc.paris-saclay.fr/)

Tip: Start **antiSMASH** and **PHASTER** first (they may take the longest). While they run, do CARD/ResFinder/PathogenFinder/VirulenceFinder/CRISPRCasFinder.

Part 1 — AMR genes (CARD vs ResFinder)

1A) CARD (RGI)

Record 2–4 of the most important AMR hits (genes or mechanisms). If you have many hits, choose the ones with the clearest clinical relevance.

AAC(6')-Iaa, aminoglycoside antibiotic, antibiotic inactivation
acrB, fluoroquinolone antibiotic, antibiotic efflux
ArnT, aminoglycoside antibiotic, antibiotic efflux

1A-ii. What does CARD say about confidence/criteria for your top hit (Perfect/Strict/Loose or similar)?

Perfect

Upload screenshot: CARD/RGI summary table (your results)

Choose File CARD_results.jpeg

Results (all Loose hits shown)

Search:

RGI Criteria	ARO Term	SNP	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence	AST Source
Perfect	AAC(6')-Iaa		protein homolog model	AAC(6',)	aminoglycoside antibiotic	antibiotic inactivation	100.0	100.00	
Strict	acrB		protein homolog model	resistance-nodulation-cell division (RND) antibiotic efflux pump	fluoroquinolone antibiotic, cephalosporin, glycylcycline, penicillin beta-lactam, tetracycline antibiotic, rifamycin antibiotic, phenicol antibiotic, disinfecting agents and antiseptics	antibiotic efflux	94.66	100.00	
Strict	ArnT		protein homolog model	pmr phosphoethanolamine transferase	peptide antibiotic	antibiotic target alteration	62.79	99.46	
Strict	bacA		protein homolog model	undecaprenyl pyrophosphate related proteins	peptide antibiotic	antibiotic target alteration	97.07	100.00	
Strict	baeR		protein overexpression model	resistance-nodulation-cell division (RND) antibiotic efflux pump	aminoglycoside antibiotic, aminocoumarin antibiotic	antibiotic efflux	94.14	100.00	

1B) ResFinder

List ResFinder AMR gene hits (or write “none detected”):

aac(6')-Iaa

Upload screenshot: ResFinder results table

Choose File ResFinder_results.jpeg

Phenotypes Hide

Other All

Antimicrobial	Class	WGS-predicted phenotype	Genetic background
gentamicin	aminoglycoside	No resistance	
tobramycin	aminoglycoside	Resistant	aac(6')-Iaa
streptomycin	aminoglycoside	No resistance	
amikacin	aminoglycoside	Resistant	aac(6')-Iaa
isepamicin	aminoglycoside	No resistance	
dibekacin	aminoglycoside	No resistance	
kanamycin	aminoglycoside	No resistance	
neomycin	aminoglycoside	No resistance	
lividomycin	aminoglycoside	No resistance	
paromomycin	aminoglycoside	No resistance	
ribostamycin	aminoglycoside	No resistance	

1C) Compare CARD vs ResFinder (interpretation)

1C-i. Do CARD and ResFinder agree on the same major AMR story? (1–2 sentences)

Yes, both CARD and ResFinder find that the aac(6')-Iaa gene is the major antibiotic resistance story.

1C-ii. If CARD reports more hits than ResFinder, what kind of hits are these likely to be? (1–2 sentences)

I can see that CARD goes into detail about how each gene contributes to antibiotic resistance, whereas ResFinder gives back basic information about the gene, such as if the gene is resistant or not. CARD, however, goes into detail about the gene itself, what the function of the gene is, and how it combats antibiotic

Part 2 — Pathogenic potential + virulence genes

2A) PathogenFinder 2

2A-i. What is the predicted outcome (probability/label) for your genome?

0.9756 probability of being a human pathogen

Upload screenshot: PathogenFinder 2 result page

Choose File PathogenFi…r_results.jpeg

Bacterial Pathogenic Capacity prediction

The pathogenic capacity of each bacterial genome is estimated using an ensemble of four neural networks, each trained on a different subset of the original dataset. The final prediction is computed as the mean output of these models. Values above 0.5 indicate pathogenic capacity, whereas values below this threshold indicate the absence of such capacity. Scores close to 0.5 reflect samples that lie near the model's decision boundary, meaning the model identifies mixed or ambiguous features that do not strongly support either class.

Module	Prediction
Neural Network 1	0.978
Neural Network 2	0.959
Neural Network 3	0.9775
Neural Network 4	0.9878
Mean	**0.9756 (0.0104)**

The bacterial genome is predicted to have *Human Pathogenic* capacity

2A-ii. One reason PathogenFinder could be wrong for your genome (1 sentence):

PathogenFinder could be wrong about my genome is that if the values are near 0.5, the sample could be mixed with other bacteria or contaminated in some way.

2B) VirulenceFinder (CGE)

List 2–6 virulence-associated genes/systems detected (or "none"):

tfaE (Prophage tail fiber assembly protein)
NCJPFIPA_03570 (IS66 family transposase ISSfl3)
apxIB (Toxin RTX-I translocation ATP-binding protein)

Upload screenshot: VirulenceFinder results table

Choose File ABRicate_s…notepad.png

#FILE	SEQUENCE	START	END	STRAND	GENE	COVERAGE	COVERAGE_MAP	GAPS	%COVERAGE	%IDENTITY	DATABASE	ACCESSION	PRODUCT
Prokka_on_dataset_31__gbk	NCJPFIPA_1	165362	165763	+	spiC/ssaB	1-402/402	==========	0/0	100.00	100.00	vfdb	NP_460358	
Prokka_on_dataset_31__gbk	NCJPFIPA_1	165765	167258	+	ssaC	1-1494/1494	==========	0/0	100.00	100.00	vfdb	NP_460359	(ssaC)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	167239	168450	+	ssaD	1-1212/1212	==========	0/0	100.00	100.00	vfdb	NP_460360	(ssaD)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	168458	168700	+	ssaE	1-243/243	==========	0/0	100.00	99.59	vfdb	NP_460361	(ssaE)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	168964	169290	+	sseA	1-327/327	==========	0/0	100.00	100.00	vfdb	NP_460362	(sseA)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	169297	169887	+	sseB	1-591/591	==========	0/0	100.00	100.00	vfdb	NP_460363	(sseB)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	169884	170357	+	sscA	1-474/474	==========	0/0	100.00	100.00	vfdb	NP_460364	(sscA)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	170360	171814	+	sseC	1-1455/1455	==========	0/0	100.00	100.00	vfdb	NP_460365	(sseC)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	171830	172417	+	sseD	1-588/588	==========	0/0	100.00	99.83	vfdb	NP_460366	(sseD)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	172420	172836	+	sseE	1-417/417	==========	0/0	100.00	100.00	vfdb	NP_460367	(sseE)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	172888	173322	+	sscB	1-435/435	==========	0/0	100.00	100.00	vfdb	NP_460368	(sscB)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	173338	174120	+	sseF	1-783/783	==========	0/0	100.00	100.00	vfdb	NP_460369	(sseF)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	174117	174806	+	sseG	1-690/690	==========	0/0	100.00	100.00	vfdb	NP_460370	(sseG)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	174900	175115	+	ssaG	1-216/216	==========	0/0	100.00	100.00	vfdb	NP_460371	(ssaG)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	175156	175383	+	ssaH	1-228/228	==========	0/0	100.00	100.00	vfdb	NP_460372	(ssaH)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	175395	175643	+	ssaI	1-249/249	==========	0/0	100.00	100.00	vfdb	NP_460373	(ssaI)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	175640	176389	+	ssaJ	1-750/750	==========	0/0	100.00	100.00	vfdb	NP_460374	(ssaJ)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	176952	177626	+	ssaK	1-675/675	==========	0/0	100.00	100.00	vfdb	NP_460376	(ssaK)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	177592	178608	+	ssaL	1-1017/1017	==========	0/0	100.00	100.00	vfdb	NP_460377	(ssaL)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	178666	179034	+	ssaM	1-369/369	==========	0/0	100.00	100.00	vfdb	NP_460378	(ssaM)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	179019	181064	+	ssaV	1-2046/2046	==========	0/0	100.00	100.00	vfdb	NP_460379	(ssaV)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	181054	182355	+	ssaN	1-1302/1302	==========	0/0	100.00	100.00	vfdb	NP_460380	(ssaN)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	182358	182735	+	ssaO	1-378/378	==========	0/0	100.00	100.00	vfdb	NP_460381	(ssaO)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	182716	183090	+	ssaP	1-375/375	==========	0/0	100.00	100.00	vfdb	NP_460382	(ssaP)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	183071	184039	+	ssaQ	1-969/969	==========	0/0	100.00	100.00	vfdb	NP_460383	(ssaQ)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	184107	184754	+	ssaR	1-648/648	==========	0/0	100.00	100.00	vfdb	NP_460384	(ssaR)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	184751	185017	+	ssaS	1-267/267	==========	0/0	100.00	100.00	vfdb	NP_460385	(ssaS)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	185018	185797	+	ssaT	1-780/780	==========	0/0	100.00	100.00	vfdb	NP_460386	(ssaT)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	185794	186852	+	ssaU	1-1059/1059	==========	0/0	100.00	100.00	vfdb	NP_460387	(ssaU)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	354291	354923	-	steA	1-633/633	==========	0/0	100.00	100.00	vfdb	NP_460542	(steA)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	376968	377918	+	sifB	1-951/951	==========	0/0	100.00	100.00	vfdb	NP_460561	(sifB)
Prokka_on_dataset_31__gbk	NCJPFIPA_1	404390	404791	+	steB	1-402/402	==========	0/0	100.00	100.00	vfdb	AAL20547	(steB)

2C) Integrate: PathogenFinder vs VirulenceFinder

2C-i. Do the virulence hits support the PathogenFinder prediction? Why or why not? (2–3 sentences)

The virulence hits do support the PathogenFinder predictions because there are a lot of genes within this sample that suggest its pathogenicity to humans. In the virulence finder via Galaxy (ABRicate), we found coding genes for secretion proteins, transposons, and prophages.

2C-ii. Give one cautious, evidence-based statement you can make about pathogenic potential from these two tools (1–2 sentences).

Based on PathogenFinder and the virulence finder though Galaxy, we can go forth with the presumption that this sample is pathogenic towards humans with active genes for prophages, transposons, and secretion proteins.

Part 3 — Secondary metabolites (antiSMASH)

3A. How many biosynthetic gene clusters (BGCs) were predicted?

3

3B. List the predicted BGC types (e.g., NRPS, PKS, bacteriocin, siderophore, terpene):

NRPS, terpene, and RiPP

3C. Choose one cluster: what might it produce / why is it interesting? (2–3 sentences)

In choosing the RiPP cluster, I found that it only had 16 functional genes. Of those 16 genes, 11 of them had "other" functions. The other 5 genes had some kind of biosynthetic or transport function.

Upload screenshot: antiSMASH overview page (showing clusters)

Choose File antiSMASH…_results.jpeg

Select genomic region:

Overview 4.1 4.2 25.1

Identified secondary metabolite regions using strictness 'relaxed'

Region	Type	From	To	Similarity Confidence	Most similar known cluster	
Region 4.1	NRP-metallophore, NRPS	1	47,598	High	enterobactin	NRPS:Type I
Region 4.2	terpene-precursor	185,644	206,543			
Region 25.1	azole-containing-RiPP	1	22,082			

No secondary metabolite regions were found in these records:

NODE_1_length_537432_cov_35.800635
NODE_2_length_491126_cov_48.127761
NODE_3_length_324196_cov_49.734900
NODE_5_length_270706_cov_40.100540
NODE_6_length_246866_cov_53.728760
NODE_7_length_223804_cov_48.030028
NODE_8_length_193475_cov_51.320008
NODE_9_length_178322_cov_42.223288
NODE_10_length_152467_cov_43.649984
NODE_11_length_149553_cov_52.185515
NODE_12_length_149367_cov_40.297267
NODE_13_length_145762_cov_41.868332
NODE_14_length_129564_cov_55.338605
NODE_15_length_114965_cov_48.488641
NODE_16_length_90516_cov_53.089541
NODE_17_length_90287_cov_38.229398
NODE_18_length_84083_cov_38.440492
NODE_19_length_82194_cov_37.819258
NODE_20_length_81337_cov_53.808959
NODE_21_length_75810_cov_44.194922
NODE_22_length_61861_cov_36.827739
NODE_23_length_58206_cov_45.835057
NODE_24_length_54283_cov_49.699443
NODE_26_length_47142_cov_43.242282
NODE_27_length_44707_cov_49.446426
NODE_28_length_39867_cov_38.052475
NODE_29_length_35616_cov_51.484172
NODE_30_length_32852_cov_51.395088
NODE_31_length_26152_cov_52.007210
NODE_32_length_26070_cov_37.212211
NODE_33_length_22449_cov_40.198507
NODE_34_length_22085_cov_35.104689
NODE_35_length_21675_cov_39.985415
NODE_36_length_21258_cov_44.231670
NODE_37_length_14343_cov_45.664657
NODE_38_length_11437_cov_42.080018
NODE_39_length_11021_cov_36.747167
NODE_40_length_7621_cov_45.672190
NODE_41_length_6709_cov_46.828709
NODE_42_length_6142_cov_109.056554
NODE_43_length_5691_cov_51.673317
NODE_44_length_5632_cov_42.003600
NODE_45_length_4188_cov_53.807346
NODE_46_length_4179_cov_38.176743
NODE_47_length_2844_cov_47.426093
NODE_48_length_2684_cov_85.985424
NODE_49_length_2608_cov_47.468985
NODE_50_length_2488_cov_89.138117
NODE_51_length_2241_cov_88.750000
NODE_52_length_2205_cov_44.525846
NODE_53_length_1749_cov_386.573565
NODE_54_length_1746_cov_81.781306
NODE_55_length_1687_cov_43.067702
NODE_56_length_1670_cov_388.281858
NODE_57_length_1649_cov_82.947201
NODE_58_length_1523_cov_152.094053
NODE_59_length_1163_cov_106.244936
NODE_60_length_1120_cov_36.060403

Part 4 — Prophages (PHASTER)

4A. How many prophage regions were reported (intact/questionable/incomplete)?

12 total prophage regions were found. Of the 12, 2 were intact, 8 regions were incomplete, and the last 2 were questionable.

4B. One sentence: what does finding prophage sequences suggest about this genome?

Finding the prophage sequences of this genome suggests that more research needs to be done to complete the unfinished sequences.

Upload screenshot: PHASTER results summary

Choose File PHASTER_results.jpeg

gi|00000000|ref|NC_000000| Genome; Raw sequence 4797060, gc%: 52.14%

Download summary as .txt file: summary.txt

Total: 12 prophage regions have been identified, of which 2 regions are intact, 8 regions are incomplete, and 2 regions are questionable.

Region	Region Length	Completeness	Score	# Total Proteins	Region Position	Most Common Phage	GC %	Details
NODE_2_length_491126_cov_48.127761								
1	13.9Kb	incomplete	20	14	399834-413791	PHAGE_Escher_RCS47_NC_042128(4)	54.59%	Show
NODE_4_length_303294_cov_44.026456								
2	6.3Kb	incomplete	30	10	52948-59310	PHAGE_Shigel_SfIV_NC_022749(2)	45.86%	Show
NODE_5_length_270706_cov_40.100540								
3	27Kb	incomplete	40	28	1-27045	PHAGE_Salmon_118970_sal3_NC_031940(27)	49.36%	Show
4	12.1Kb	incomplete	40	16	258523-270706	PHAGE_Entero_SfV_NC_003444(4)	46.18%	Show
NODE_8_length_193475_cov_51.320008								
5	20.4Kb	incomplete	60	24	18369-38788	PHAGE_Burkho_BcepMu_NC_005882(14)	50.58%	Show
NODE_9_length_178322_cov_42.223288								
6	7.4Kb	incomplete	20	8	131003-138406	PHAGE_Escher_500465_2_NC_049343(4)	53.09%	Show
NODE_15_length_114965_cov_48.488641								
7	40.5Kb	questionable	70	62	309-40825	PHAGE_Entero_UAB_Phi20_NC_031019(36)	47.00%	Show
NODE_19_length_82194_cov_37.819258								
8	15.3Kb	questionable	70	14	1-15357	PHAGE_Salmon_Fels_1_NC_010391(3)	51.39%	Show
NODE_21_length_75810_cov_44.194922								
9	28.5Kb	incomplete	60	17	15474-43994	PHAGE_Escher_500465_1_NC_049342(12)	54.80%	Show
NODE_34_length_22085_cov_35.104689								
10	11.5Kb	incomplete	60	9	82-11632	PHAGE_Entero_HK630_NC_019723(2)	43.38%	Show
NODE_35_length_21675_cov_39.985415								
12	14.3Kb	intact	150	17	1-14342	PHAGE_Phage_Gifsy_1_NC_010392(17)	53.99%	Show

PHASTER New Search Genomes Help About My Sear

Intact (score > 90)
Questionable (score 70-90)
Incomplete (score < 70)

Region: The number assigned to the region.
Region Length: The length of the sequence of that region (in bp).
Completeness: A prediction of whether the region contains a intact or incomplete prophage based on the above criteria.
Score: The score of the region based on the above criteria.
 # Total Proteins: The number of ORFs present in the region.
Region Position: The start and end positions of the region on the bacterial chromosome.
Most Common Phage: The phage(s) with the highest number of proteins most similar to those in the region.
GC %: The percentage of GC nucleotides of the region.

Part 5 — Genomic islands (IslandViewer)

This software will require your **.gbk** file from Prokka.

5A. Roughly how many islands are predicted (or what is the overall pattern)?

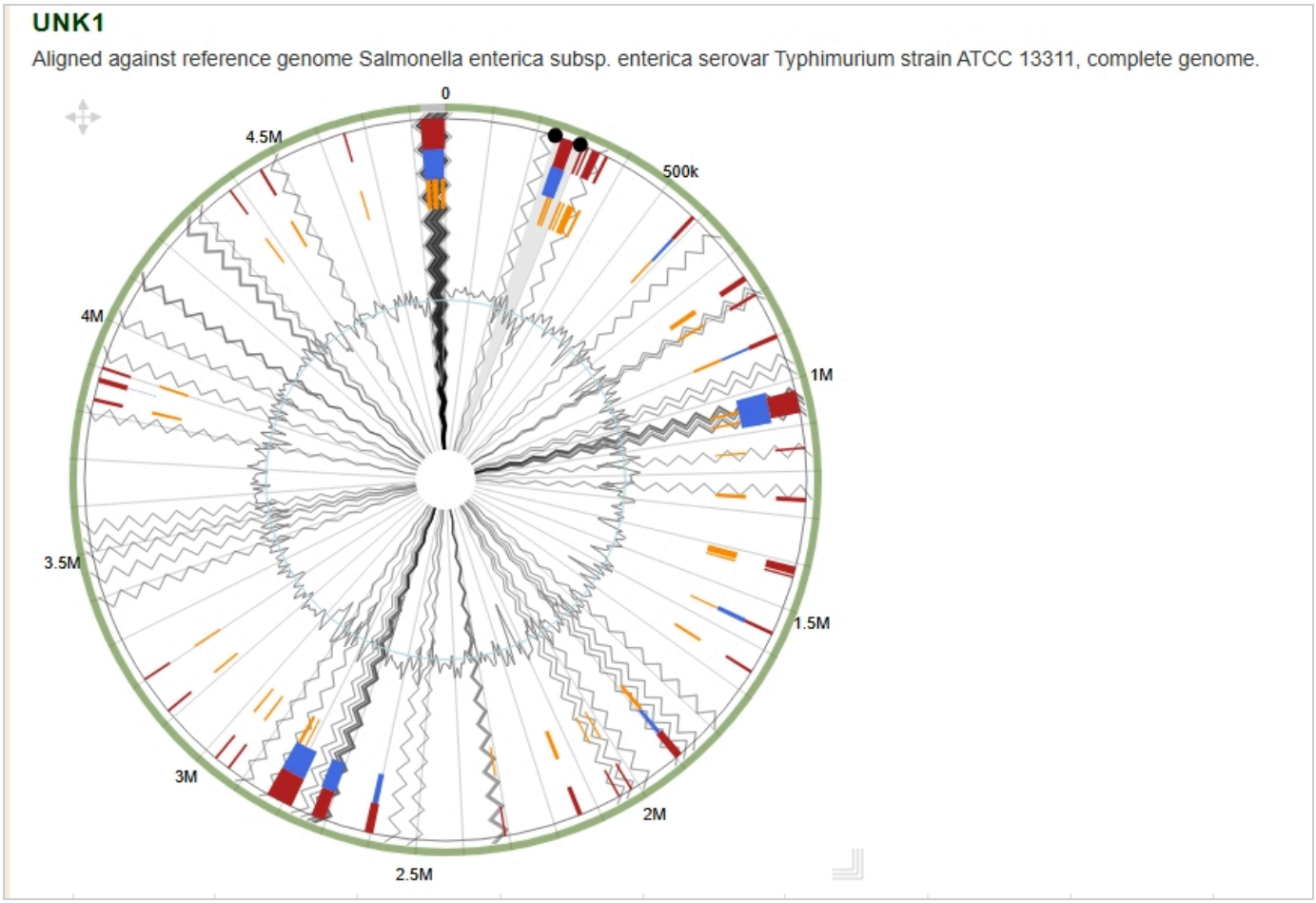
There are about 10 predicted islands of varying sizes. There are 4 bigger islands and 6 smaller islands.

5B. Pick one island: what kind of genes does it appear enriched for? (AMR/virulence/metabolism/unknown) (1–2 sentences)

I picked one of the larger islands and it showed me genes that coded for metabolism-related functions, such as enzyme maturation and protein transport.

Upload screenshot: IslandViewer genome map / island overview

Choose File IslandViewe…eenshot.jpeg



Part 6 — CRISPR-Cas systems (CRISPRCasFinder)

6A. Does CRISPRCasFinder detect CRISPR arrays? If yes, how many?

5 arrays

6B. Does it detect Cas genes / a Cas system type or subtype? Record what it reports.

Yes, CRISPR/CasFinder found 2 Cas arrays. One of the Cas clusters was a TypeIE and the other was simply labeled as CAS.

6C. One sentence interpretation: what does CRISPR presence suggest about this bacterium's history?

CRISPR presence suggests that this bacterium's history contains gene editing machinery, which can either be used to edit its own genome or humans can extract that gene and implement that gene editing process into

Upload screenshot: CRISPRCasFinder results (arrays + Cas summary)

Choose File Screenshot…-saclay.fr.jpeg

Sequence NODE_2_length_491126_cov_48_127761 [491126 bp] [4 CRISPR] [21 spacers] [12 evidence level]

Summary Details

Element	CRISPR Id / Cas Type	Start	End	Spacer / Gene	Repeat consensus / cas genes	Direction	Evidence Level
CRISPR	NODE_2_length_491126_cov_48_127761_2	367337	367609	4	GTGTTCCCCGCGCCAGCGGGGATAAACCG	+	3
Cas cluster	CAS-TypeIE	375185	383644	8	cas3_TypeI, cse1_TypeIE, cse2_TypeIE, cas7_TypeIE, cas5_TypeIE, cas6_TypeIE, cas1_TypeIE, cas2_TypeIE		
CRISPR	NODE_2_length_491126_cov_48_127761_3	383741	384199	7	GTGTTCCCCGCGCCAGCGGGGATAAACCG	+	4
CRISPR	NODE_2_length_491126_cov_48_127761_4	384273	384850	9	GTGTTCCCCGCGCCAGCGGGGATAAACCG	+	4

Part 7 — "Genome story" synthesis

7A. Pick the three most interesting findings across ALL tools (AMR, virulence/pathogen potential, BGCs, prophages, islands, CRISPR). For each: state (i) the evidence and (ii) one cautious interpretation.

Some of the most interesting finds that I came across doing this assignment was the pathogenic potential from CARD, navigating and finding the virulence factors in the IslandViewer map, and assessing the CRISPR/Cas genes from the CRISPR/CasFinder. With CARD, the I was able to find a lot of genes that gave my sample a high pathogenicity rating when I took the sample to PathogenFinder. With that information in mind, when I went to the IslandViewer, I was looking at the different islands and found multiple genes relating to bacterial pathogenicity and particular virulence factors. Lastly, when I went onto the CRISPR/CasFinder, I found 5

7B. One thing you would want to test experimentally (or with additional data) based on today's results (1–2 sentences).

genome to other genomes. I feel like if theis bacterium has a high chance of being pathogenic with various virulence factors, I believe it would be a good idea to investigate its ability to spread its antibiotic resistance.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
sampleID: Unk1
assembly file: Unk1_SPAdes_filtered_contigs.fasta
suspected identity: Salmonella enterica subsp. enterica serovar Typhimurium

AMR (CARD)
Top hits:

```
AAC(6')-Iaa, aminoglycoside antibiotic, antibiotic inactivation
acrB, fluoroquinolone antibiotic, antibiotic efflux
ArnT, aminoglycoside antibiotic, antibiotic efflux
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Confidence/criteria: Perfect

AMR (ResFinder)

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aac(6')-Iaa
```

Compare: Yes, both CARD and ResFinder find that the aac(6')-Iaa gene is the major antibiotic resistance story.
If CARD reports more: I can see that CARD goes into detail about how each gene contributes to antibiotic resistance, whereas ResFinder gives back basic information about the gene, such as if the gene is resistant or not. CARD, however, goes into detail about the gene itself, what the function of the gene is, and how it combats antibiotic medication/treatment.

Pathogenicity + virulence
PathogenFinder: 0.9756 probability of being a human pathogen
PathogenFinder limitation: PathogenFinder could be wrong about my genome is that if the values are near 0.5, the sample could be mixed with other bacteria or contaminated in some way.
VirulenceFinder hits:

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tfaE (Prophage tail fiber assembly protein)
NCJPFIPA_03570 (IS66 family transposase ISSfl3)
apxIB (Toxin RTX-I translocation ATP-binding protein)
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Compare (VF vs PF): The virulence hits do support the PathogenFinder predictions because there are a lot of genes within this sample that suggest its pathogenicity to humans. In the virulence finder via Galaxy (ABRicate), we found coding genes for secretion proteins, transposons, and prophages.
Cautious claim: Based on PathogenFinder and the virulence finder though Galaxy, we can go forth with the presumption that this sample is pathogenic towards humans with active genes for prophages, transposons, and secretion proteins.

antiSMASH
 # BGCs: 3
Types: NRPS, terpene, and RiPP
One cluster interpretation: In choosing the RiPP cluster, I found that it only had 16 functional genes. Of those

16 genes, 11 of them had "other" functions. The other 5 genes had some kind of biosynthetic or transport function.

PHASTER
Counts: 12 total prophage regions were found. Of the 12, 2 were intact, 8 regions were incomplete, and the last 2 were questionable.
Interpretation: Finding the prophage sequences of this genome suggests that more research needs to be done to complete the unfinished sequences.

IslandViewer
Pattern/# islands: There are about 10 predicted islands of varying sizes. There are 4 bigger islands and 6 smaller islands.
One island interpretation: I picked one of the larger islands and it showed me genes that coded for metabolism-related functions, such as enzyme maturation and protein transport.

CRISPRCasFinder
Arrays: 5 arrays
Cas system: Yes, CRISPR/CasFinder found 2 Cas arrays. One of the Cas clusters was a TypeIE and the other was simply labeled as CAS.
Interpretation: CRISPR presence suggests that this bacterium's history contains gene editing machinery, which can either be used to edit its own genome or humans can extract that gene and implement that gene editing process into other organisms.

Genome story
Three most interesting findings:

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Some of the most interesting finds that I came across doing this assignment was the pathogeni
c potential from CARD, navigating and finding the virulence factors in the IslandViewer map,
and assessing the CRISPR/Cas genes from the CRISPR/CasFinder. With CARD, the I was able to fi
nd a lot of genes that gave my sample a high pathogenicity rating when I took the sample to P
athogenFinder. With that information in mind, when I went to the IslandViewer, I was looking
at the different islands and found multiple genes relating to bacterial pathogenicity and par
ticular virulence factors. Lastly, when I went onto the CRISPR/CasFinder, I found 5 CRISPR ar
rays and 2 Cas systems. This evidence tells me that this bacterium has the potential to edit
genomes, whether it be its own genome or other organisms' genomes.
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Next test: One thing that I would like to test is how successful this bacterium can horizontally transfer pieces of its genome to other genomes. I feel like if theis bacterium has a high chance of being pathogenic with various virulence factors, I believe it would be a good idea to investigate its ability to spread its antibiotic resistance.